

This model analyzes the sociotechnical dynamics of a Generative AI Coding Assistant. We will be focusing on user adoption, server performance limits, and open source community effects.

Our primary stock is Active AI users which is governed by New user onboarding (inflow) and users churn rate as outflow.

Key Assumptions:

- Time Scale: The system operates on short deployment window that is measured in days/weeks.
- Pricing & subscription: As I wanted to stay on technical aspect and performance of the system, that's why subscription price is held completely constant to isolate system behaviour purely to technical performance and network effects which excludes financial ~~the~~ constraints.
- Users Contribution: Active users linearly contributes open source plugins and prompts templates, directly increasing platform utility.

• Model 1:

Onboarding and churn operate purely on fixed external rates. No feedback exists between stock and flow.

- Baseline empirical constants chosen to represent a fixed daily marketing inflow and natural user drop-off.

- Model 2:

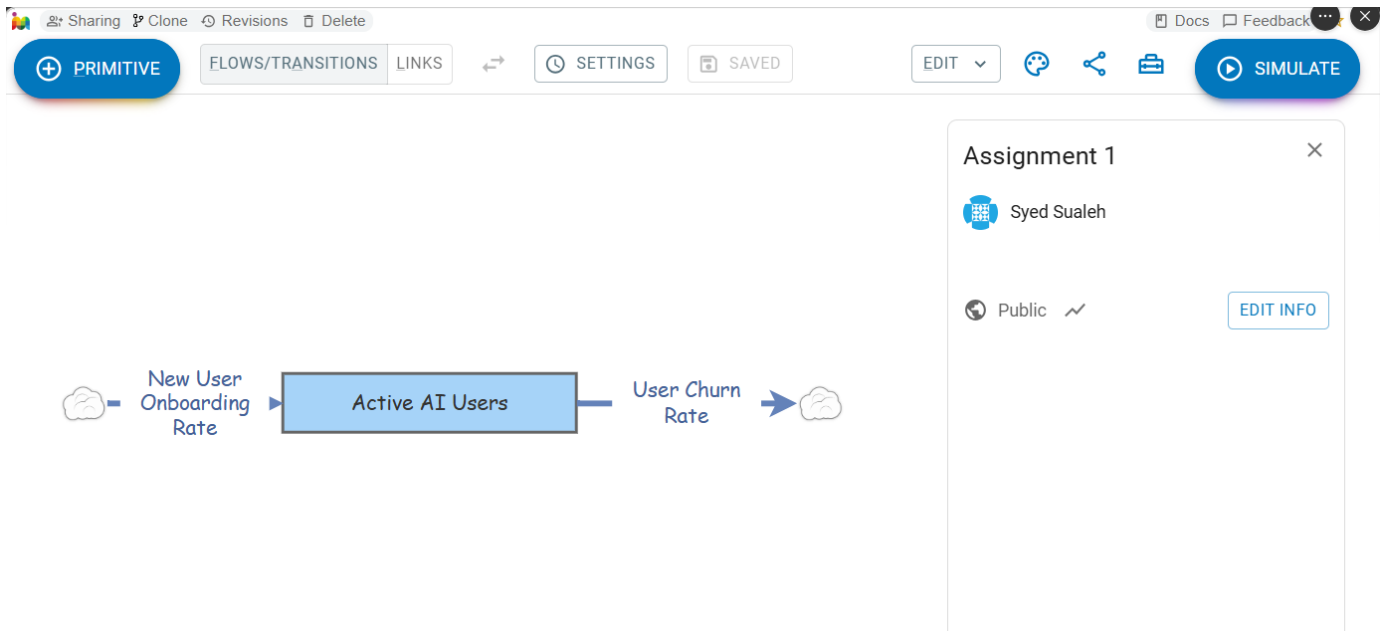
In this system, the constraints create a self-correcting negative feedback loop.

- Linear scaling factors reflect proportional system degradation. They cause the stock to naturally balance around maximum server capacity.

- Model 3:

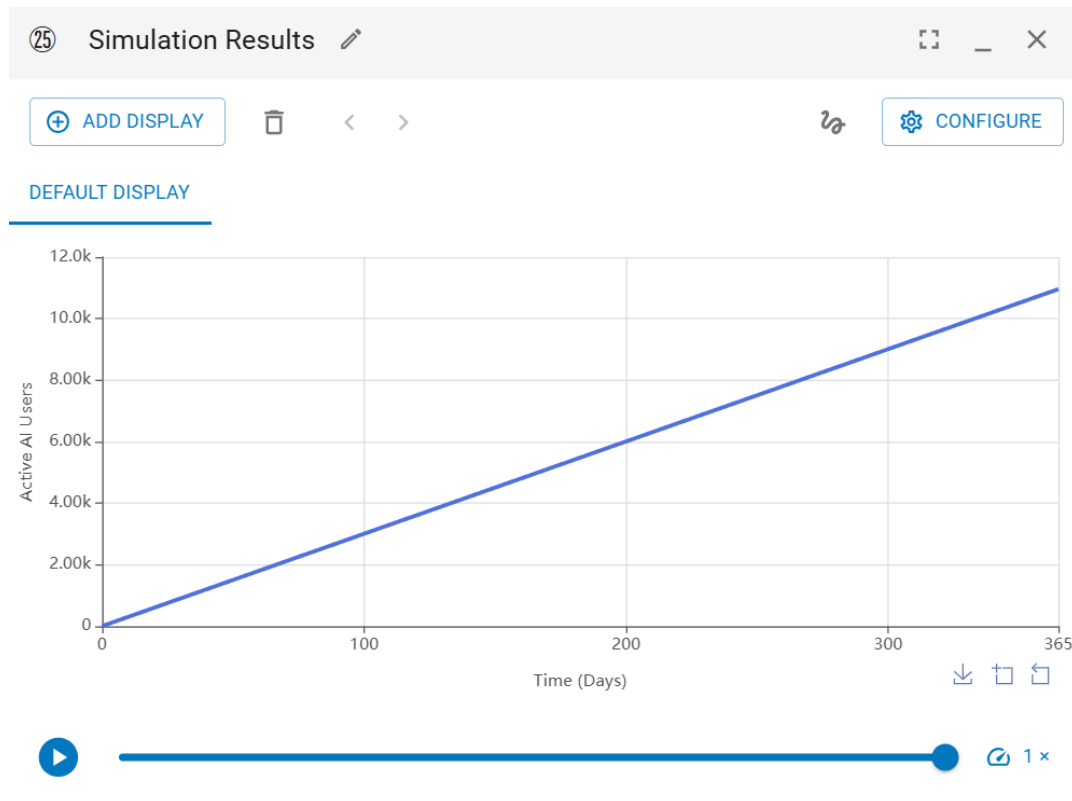
As we get active users, I assume about 10% of those build extra plugins, prompt templates, and custom add-ons to make their work easier. As it happens, the platform becomes more ~~useful~~ useful and easier with free plugins. So overall platform value goes up. more features brings more people so more people sign up. Now the cycle repeats as ~~more~~ new users create more features and plugins.

Stock flow diagram without any feedback loops:



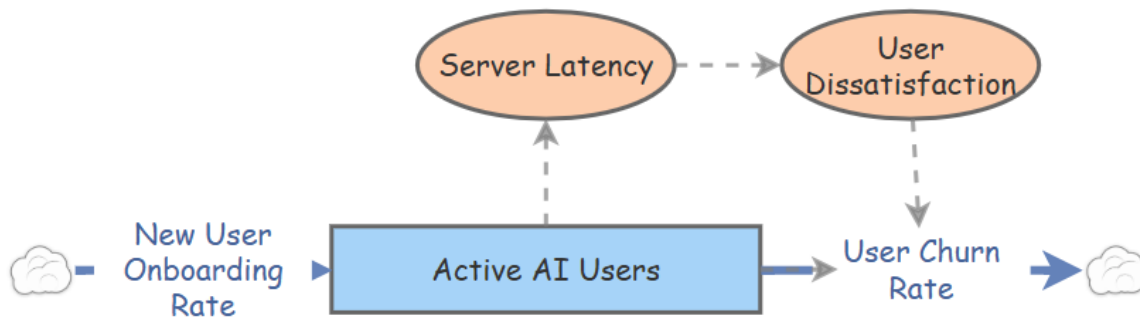
Inflow Rate 50 users/day, Outflow rate 20 users/day

Simulation Results:



Linear growth. Inflow > outflow, the stock increases in a straight diagonal line (Net = +30 users/day).

Stock-and-flow diagram with only balancing loops:



Server latency=[Active AI Users]*0.08

User dissatisfaction=[Server Latency]*0.2

Churn rate=[Active AI Users]*0.02+[User Dissatisfaction]*2

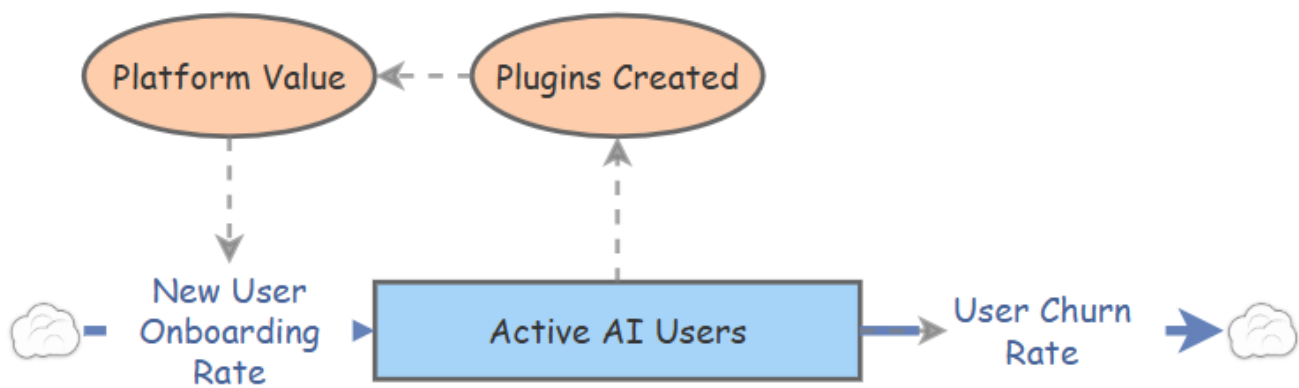
Onboarding rate= 100 users/day

Simulation Results:



The active user count follows an curve growth path, rapidly increasing before leveling off and becoming straight near 1,900; server latency increases dissatisfaction and churn.

Stock-and-flow diagram with only reinforcing loops.



Churn rate=10

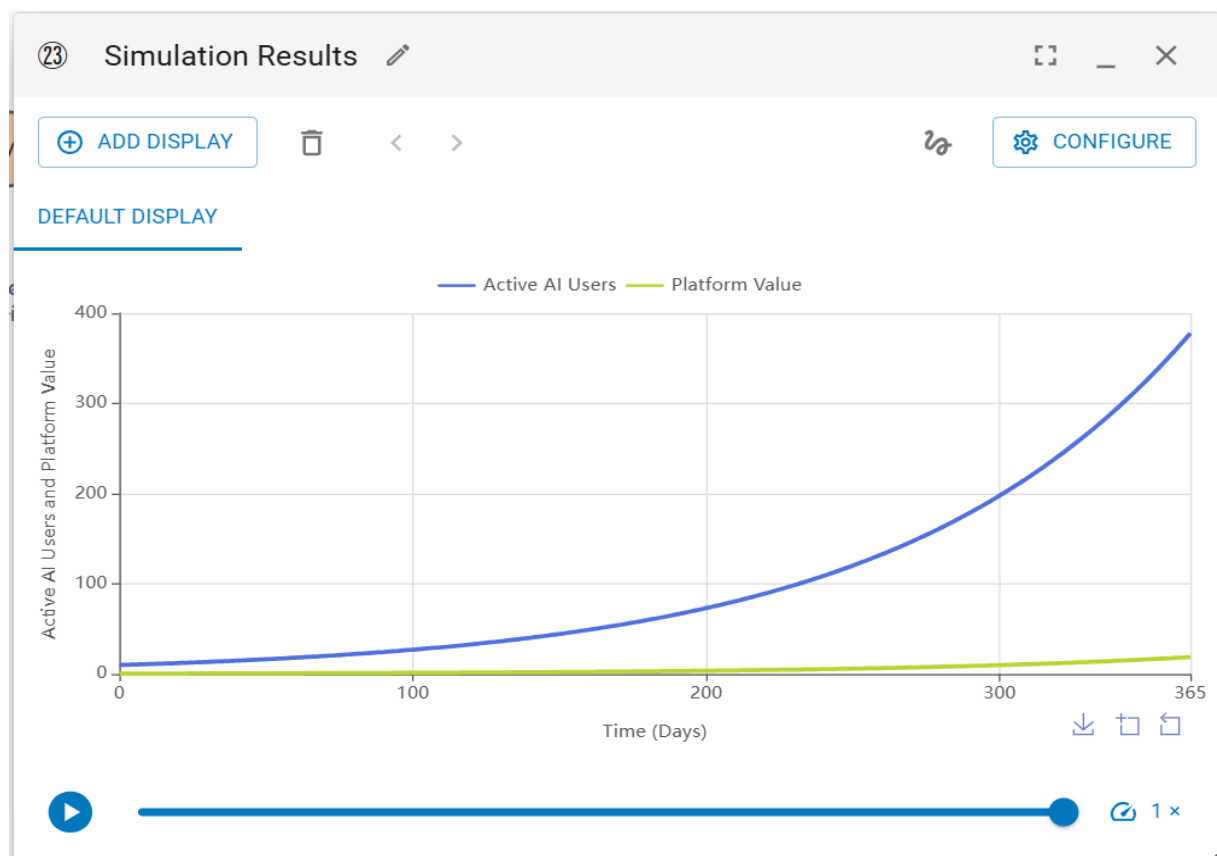
Initial value of active ai users=10

Plugins=[Active AI Users]*0.1

Platform value=[Plugins Created]*0.5

New onboarding rate=10+([Platform Value]*0.2)

Simulation results:



The active user count follows an accelerating upward exponential curve effected by community plugin creation and compounding platform value.

Submission by:

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D-3